The descent set of the superstandard tableau determines the dominance order of the shape

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Editorial note. The counted statement is the author's labelled lemma 'lem:superstandard characterize coefficients' (source lines 570-577); the proof environment begins after two blank lines, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence introducing the statement.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source, which already carries the author's theorem-environment family and macro set; packages needed only by other parts of the article are left out because no retained line uses them.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the characterisation of the superstandard tableau by its descent set (author label lem:superstandard characterize coefficients).

Allowed standard prerequisites (editorial).

- Superstandard tableau T_lambda. T_lambda is the standard Young tableau of shape lambda filled row by row with 1,2,...,N.. Both assertions are about T_lambda and its descent set.
- Descent set and conjugate partition. Des(T) is the set of positions i with i+1 in a lower row than i, and lambda' denotes the conjugate partition.. The statement identifies Des(T_lambda) with [N-1] minus the partial sums of lambda', and the proof argues by columns of the conjugate shape.

The power of these notions may be reflected by the following statement:

Lemma 0.1. Let $\lambda, \mu \vdash N$ be two partitions. Then:

1. $\mathrm{Des}(T_\lambda) = [N-1] \setminus \{\lambda'_1, \lambda'_1 + \lambda'_2, \dots\}$.
2. If $\mathrm{Des}(T) = [N-1] \setminus \{\lambda'_1, \lambda'_1 + \lambda'_2, \dots\}$ for some $T \in \mathrm{SYT}(\mu)$, then $\mu' \geq \lambda'$. Furthermore, if $\mu = \lambda$ then $T = T_\lambda$.

Proof. The first assertion is obvious, so let us focus on the second assertion.

Let $T \in \mathrm{SYT}(\mu)$ be a tableau, and assume that $\mathrm{Des}(T) = \mathrm{Des}(T_\lambda)$ and $T \neq T_\lambda$. We aim to show that $\mu' > \lambda'$. Let us denote the first column that differs between T and T_λ by i. The i-th column of T_λ contains the entries $s+1, \dots, s+\lambda'_i$, where $s = \lambda'_1 + \cdots + \lambda'_{i-1}$. To prove $\mu' > \lambda'$, it suffices to show that all these entries also appear in the i-th column of T.

Assume, by contradiction, that some of these entries appear in other columns of T. Let x be the minimal such entry, and let $j \neq i$ be the column of T that contains x. Since $s+1$ appears in the i-th column of T, we may assume that $x > s+1$. If $j < i$, this contradicts the assumption that T and T_λ agree in the first $i-1$ columns. On the other hand, if $j > i$, then x appears in the first row of T, since it is the minimal entry of the j-th column of T. Consequently, we obtain that $x-1 \notin \mathrm{Des}(T)$, while $x-1 \in \mathrm{Des}(T_\lambda)$, which contradicts the assumption that $\mathrm{Des}(T) = \mathrm{Des}(T_\lambda)$. $\square$